

Log-Square Horizons and the Square-Root Block-Contraction Law

A Sufficient Dyadic Criterion for Polylogarithmic Directional Hardy Energy

Liang Wang

July 2026

Abstract

RH-74 closed the folded-Gaussian production chain at five fixed noise scales. The remaining Stage A1 problem is no longer finite-scale validation but a uniform theorem over the small-noise dyadic family. This paper isolates a sharp sufficient scaling law for that theorem.

Let $\sigma_k = \sigma_0 2^{-k}$ and let (A_k, X_k, Y_k) be a directional bulk triple. Suppose there are block horizons $M_k \leq (k + a)^2$ such that

$$\left\| A_k^{M_k} \right\|_2 \leq C_q \sqrt{\sigma_k}, \quad \sigma_k \|Y_k\|_F^2 \leq C_y,$$

and suppose the one-block source energy and finite transfer prefix obey

$$\sum_{r < M_k} \|A_k^r X_k\|_F^2 \leq C_s (k + a)^s, \text{quad} \sum_{r < M_k} \|Y_k A_k^r X_k\|_F^2 \leq C_f (k + a)^f.$$

Then the infinite block tail satisfies

$$T_k^2 \leq \frac{C_y C_q^2 C_s}{1 - C_q^2 \sigma_0} (k + a)^s,$$

and the full Hardy energy squared is $O((k + a)^{\max\{s, f\}})$. Thus a log-square growing horizon is fully compatible with the required polylogarithmic Stage A1 input. The $\sqrt{\sigma_k}$ factor is the exact scale that cancels the $\|Y_k\|_F^2 = O(\sigma_k^{-1})$ mesh growth.

A 256-bit Arb audit combines the analytic-upstream block contractions of RH-74 with RH-70's source, observation, prefix, and tail balls. At the five levels $k = 0, \dots, 4$, all ten channels satisfy common constants

$$a = 2, \text{quad} C_q = 0.086, \text{quad} C_y = 2.561, \text{quad} C_s = 3.1, \text{quads} = 0, \text{quad} C_f = 0.552, \text{quad} f = 1.$$

The common tail envelope is 0.058788, while the largest certified tail is 0.049421.

This identifies the correct all-level target but does not prove it beyond the five anchors. The next open mechanism is phase compression or effective-rank decay sufficient to imply the square-root block law. Stage A1, unconditional Stage A4, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: block contraction; Hardy energy; dyadic family; small noise; logarithmic horizon; interval arithmetic.

MSC 2020: 47A10; 47B35; 65G20; 93B28.

1 Introduction

The earlier directional-Hardy program deliberately separated finite matrix identities from small-noise family statements. RH-61 showed that a worst-case one-step geometric envelope may

require a power-law horizon when the norm gap closes [5]. RH-68 then proved that stability, spectral radius, and even a perfectly conditioned Lyapunov metric cannot force a universal fixed Krylov depth [2]. These are genuine barriers to fixed-depth arguments, not barriers to a controlled growing depth.

RH-70 found production block horizons

$$4, 9, 16, 25, 32$$

at dyadic noise scales 0.16, 0.08, 0.04, 0.02, 0.01 and certified the frozen tails [4]. RH-74 transported those contractions to the analytic upstream triples and closed the complete finite-scale bridge [3]. The horizon list suggests the log-square schedule $(k+2)^2$, but a schedule alone is not enough: the observation Frobenius norm grows with dimension.

The decisive empirical quantity is instead

$$\frac{\|A_k^{M_k}\|}{\sqrt{\sigma_k}}. \quad (1)$$

It remains bounded by 0.086 in every validated channel. This paper explains why (1), rather than a fixed contraction constant, is the natural uniform target.

2 Dyadic block setting

Let

$$\sigma_k = \sigma_0 2^{-k}, \quad k \geq 0,$$

and consider finite directional triples (A_k, X_k, Y_k) . Define

$$S_k(M) = \sum_{r=0}^{M-1} \|A_k^r X_k\|_{\text{F}}^2, \quad (2)$$

$$F_k(M) = \sum_{r=0}^{M-1} \|Y_k A_k^r X_k\|_{\text{F}}^2, \quad (3)$$

$$H_k^2 = \sum_{r \geq 0} \|Y_k A_k^r X_k\|_{\text{F}}^2. \quad (4)$$

The block-power theorem gives, whenever $q_k = \|A_k^M\|_{\text{F}} < 1$,

$$H_k^2 \leq F_k(M) + \|Y_k\|_{\text{F}}^2 \frac{q_k^2}{1 - q_k^2} S_k(M). \quad (5)$$

For the mesh-locked production family, the fine dimension obeys $n_k \sigma_k = 5.12$ and the observation rank is proportional to n_k . Consequently $\|Y_k\|_{\text{F}}^2 = O(\sigma_k^{-1})$. A merely uniform $q_k < q < 1$ would leave a power divergence in (5). The block contraction must improve with the mesh.

3 The square-root block law

Theorem 3.1 (Log-square square-root block criterion). *Fix constants $a > 0$, $C_q, C_y, C_s, C_f \geq 0$ and exponents $s, f \geq 0$. Suppose for every $k \geq 0$ there is an integer M_k such that*

$$M_k \leq (k + a)^2, \quad (6)$$

$$q_k := \|A_k^{M_k}\|_F \leq C_q \sqrt{\sigma_k}, \quad (7)$$

$$\sigma_k \|Y_k\|_F^2 \leq C_y, \quad (8)$$

$$S_k(M_k) \leq C_s (k + a)^s, \quad (9)$$

$$F_k(M_k) \leq C_f (k + a)^f. \quad (10)$$

If $C_q^2 \sigma_0 < 1$, then

$$H_k^2 \leq C_f (k + a)^f + \frac{C_y C_q^2 C_s}{1 - C_q^2 \sigma_0} (k + a)^s. \quad (11)$$

In particular, $H_k = O((k + a)^{\max\{s, f\}/2})$, equivalently a polylogarithmic bound in $1/\sigma_k$.

Proof. Insert (7)–(10) into (5). Since $\sigma_k \leq \sigma_0$,

$$\frac{\|Y_k\|_F^2 q_k^2}{1 - q_k^2} \leq \frac{(C_y/\sigma_k) C_q^2 \sigma_k}{1 - C_q^2 \sigma_0}.$$

Multiplication by (9) proves (11). Finally $k = \log_2(\sigma_0/\sigma_k)$. $\square$

Remark 3.2 (Why the exponent 1/2 is critical). *If $q_k = O(\sigma_k^\theta)$ while $\|Y_k\|_F^2 = O(\sigma_k^{-1})$, the raw tail prefactor scales as $\sigma_k^{2\theta-1}$. Thus $\theta = 1/2$ is the threshold between a power loss and a bounded prefactor. Additional source growth may still be polylogarithmic.*

Remark 3.3 (No fixed-depth claim). *The theorem permits $M_k = O(k^2) = O(\log^2(1/\sigma_k))$. It is therefore consistent with both RH-61’s geometric-horizon warning and RH-68’s exact fixed-depth no-go theorem.*

4 Five-anchor Arb certificate

The audit reads the exact-dyadic RH-70 balls and the analytic true-block balls from RH-74. Every derived ratio and common envelope is recomputed at 256-bit Arb precision [1].

Table 1: Selected horizon and largest left/right normalized block contraction at each dyadic level.

k	σ_k	M_k	$(k + 2)^2$	$\max_s q_{k,s}/\sqrt{\sigma_k}$
0	0.16	4	4	0.052554
1	0.08	9	9	0.055679
2	0.04	16	16	0.085680
3	0.02	25	25	0.084832
4	0.01	32	36	0.081695

The common constants are

$$a = 2, \text{quad}C_q = 0.086, \text{quad}C_y = 2.561, \text{quad}C_s = 3.1, \text{quads} = 0, \text{quad}C_f = 0.552, \text{quad}f = 1. \quad (12)$$

For these values the theorem gives the level-independent tail envelope

$$\frac{C_y C_q^2 C_s}{1 - C_q^2 \sigma_0} < 0.058788. \quad (13)$$

Table 2: Largest channel values entering the common envelope.

σ	$\sigma \ Y\ _F^2$	S_k	$F_k/(k+2)$	actual tail
0.16	2.5600	1.1565	0.5027	0.006471
0.08	2.5600	1.5997	0.5337	0.010823
0.04	2.5600	2.2047	0.5510	0.040363
0.02	2.5600	2.6711	0.5340	0.041824
0.01	2.5600	3.0993	0.5165	0.049421

All entries remain below (12); every actual tail remains below (13). The resulting conditional all-level estimate is

$$H_k^2 \leq 0.552(k+2) + 0.058788, \quad H_k = O(\sqrt{k}).$$

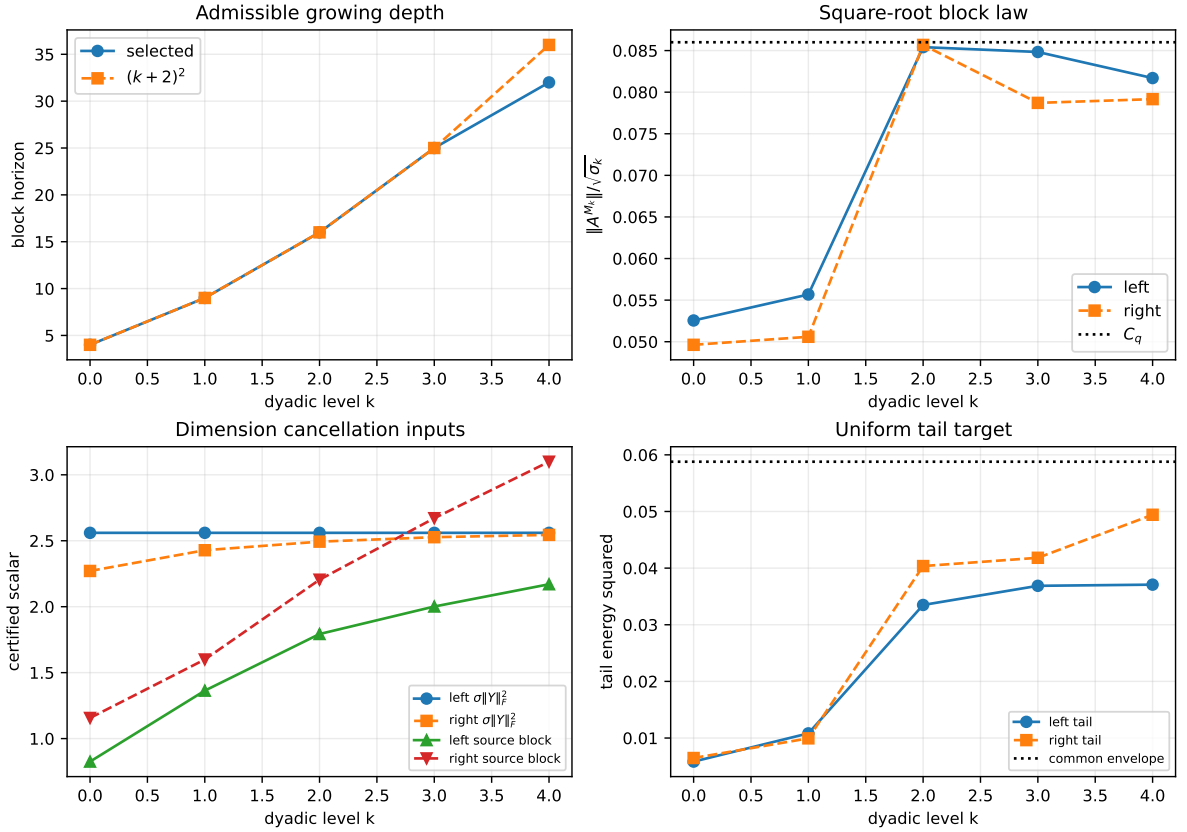


Figure 1: The log-square schedule, square-root contraction ratio, mesh-growth cancellation inputs, and common tail envelope.

5 What remains to prove

The theorem converts the vague phrase “uniform horizon control” into four specific all-level inequalities: (7)–(10). The five interval anchors show that these inequalities are compatible

with the physical production family and have nontrivial margin. They do not establish an induction in k or a continuum estimate.

The next plausible mechanisms are:

1. phase support compressed to arcs whose width shrinks along the dyadic family, giving polynomial approximation at depth $O(k^2)$;
2. weighted effective-rank decay, allowing a small exceptional phase tail;
3. a direct strong–weak block estimate for the noisy transfer operator.

RH-76 addresses the first mechanism. If phase compression fails, RH-77 will test whether effective-rank decay can replace it.

No all-dyadic law is proved here, so uniform Stage A1 and unconditional Stage A4 remain open. This paper does not construct a renormalized determinant, self-adjoint Hilbert–Polya operator, $T \log T$ counting law, prime-power trace formula, zeta-zero identity, or proof of the Riemann Hypothesis.

6 Conclusion

A universal fixed horizon is neither necessary nor expected. The correct target is a log-square horizon paired with a square-root-in-noise block contraction. That factor precisely neutralizes the observation’s mesh growth and leaves a polylogarithmic Hardy bound. Five validated levels satisfy a single conservative constant ledger. The route remains open, but its next mathematical obligation is now explicit.

References

- [1] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633.
- [2] Liang Wang. Phase-coherence barriers to uniform block krylov depth. Preprint and software repository, RH-68, 2026.
- [3] Liang Wang. Validated upstream-to-frozen hardy bridges for the folded-gaussian production chain. Preprint and software repository, RH-74, 2026.
- [4] Liang Wang. Outward-rounded block hardy certificates for frozen production matrices. Preprint and software repository, RH-70, 2026.
- [5] Liang Wang. Directional horizon scaling for phase-aware stein tails. Preprint and software repository, RH-61, 2026.